WORKED SOLUTION

# STEP III 1995, Question 4

Pure • Trigonometric Series Summation

**STEP III 1995, Question 4**

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/95-s3-q4

## The key idea

A sum of cosines is the real part of a sum of complex exponentials, and $\sum e^{\mathrm{i}k\theta}$ is a geometric series. Summing it and then pulling out a "half-angle" factor $e^{\mathrm{i}\theta/2}$ from top and bottom turns the awkward $e^{\mathrm{i}m\theta}-1$ terms into sines, which is where the $\sin(\frac{1}{2}\theta)$ in the denominator comes from. For the inequality, the trick is to rewrite $\mathrm{C}_n(\theta)$ so that $\frac{1}{2}$ visibly splits off.

## Solution

### The sums $\mathrm{C}_n$ and $\mathrm{S}_n$

Put $z = e^{\mathrm{i}\theta}$. Since $0 < \theta < 2\pi$, we have $z \neq 1$, so the geometric series formula applies:

$$\mathrm{C}_n(\theta) + \mathrm{i}\,\mathrm{S}_n(\theta) = \sum_{k=0}^{n} e^{\mathrm{i}k\theta} = \frac{e^{\mathrm{i}(n+1)\theta} - 1}{e^{\mathrm{i}\theta} - 1}.$$

Now use $e^{\mathrm{i}\phi} - 1 = e^{\mathrm{i}\phi/2}(e^{\mathrm{i}\phi/2} - e^{-\mathrm{i}\phi/2}) = 2\mathrm{i}\,e^{\mathrm{i}\phi/2}\sin(\frac{1}{2}\phi)$ on the top (with $\phi = (n+1)\theta$) and on the bottom (with $\phi = \theta$):

$$\mathrm{C}_n(\theta) + \mathrm{i}\,\mathrm{S}_n(\theta) = \frac{2\mathrm{i}\,e^{\mathrm{i}(n+1)\theta/2}\sin(\frac{1}{2}(n+1)\theta)}{2\mathrm{i}\,e^{\mathrm{i}\theta/2}\sin(\frac{1}{2}\theta)} = e^{\mathrm{i}n\theta/2}\,\frac{\sin(\frac{1}{2}(n+1)\theta)}{\sin(\frac{1}{2}\theta)}.$$

The denominator is non-zero because $0 < \frac{1}{2}\theta < \pi$. The fraction is real, so taking real and imaginary parts (with $e^{\mathrm{i}n\theta/2} = \cos\frac{1}{2}n\theta + \mathrm{i}\sin\frac{1}{2}n\theta$) gives

$$\mathrm{C}_n(\theta) = \frac{\cos(\frac{1}{2}n\theta)\sin(\frac{1}{2}(n+1)\theta)}{\sin(\frac{1}{2}\theta)}, \qquad \mathrm{S}_n(\theta) = \frac{\sin(\frac{1}{2}n\theta)\sin(\frac{1}{2}(n+1)\theta)}{\sin(\frac{1}{2}\theta)}.$$

The first is the required result, and the second is the corresponding expression for $\mathrm{S}_n$.

**Ask yourself** Is there a way to see this without complex numbers?

Yes: multiply each term by $2\sin(\frac{1}{2}\theta)$ and use the product-to-sum formula

$$2\sin(\tfrac{1}{2}\theta)\cos k\theta = \sin((k+\tfrac{1}{2})\theta) - \sin((k-\tfrac{1}{2})\theta).$$

Summing from $k = 0$ to $n$, everything telescopes except the last and first terms:

$$2\sin(\tfrac{1}{2}\theta)\,\mathrm{C}_n(\theta) = \sin((n+\tfrac{1}{2})\theta) - \sin(-\tfrac{1}{2}\theta) = \sin((n+\tfrac{1}{2})\theta) + \sin(\tfrac{1}{2}\theta). \qquad (*)$$

By the sum-to-product formula $\sin A + \sin B = 2\sin\frac{1}{2}(A+B)\cos\frac{1}{2}(A-B)$, with $A = (n+\frac{1}{2})\theta$, $B = \frac{1}{2}\theta$, the right-hand side is $2\sin(\frac{1}{2}(n+1)\theta)\cos(\frac{1}{2}n\theta)$, which agrees with what we found above.

#### The inequality

Equation $(*)$ is exactly the form we want. Dividing by $2\sin(\frac{1}{2}\theta) \neq 0$,

$$\mathrm{C}_n(\theta) = \frac{1}{2} + \frac{\sin((n+\frac{1}{2})\theta)}{2\sin(\frac{1}{2}\theta)}.$$

(Starting from the given formula instead, use $2\cos(\frac{1}{2}n\theta)\sin(\frac{1}{2}(n+1)\theta) = \sin((n+\frac{1}{2})\theta) + \sin(\frac{1}{2}\theta)$ to get the same line.) Hence

$$\left|\mathrm{C}_n(\theta) - \frac{1}{2}\right| = \frac{|\sin((n+\frac{1}{2})\theta)|}{2\sin(\frac{1}{2}\theta)} \leq \frac{1}{2\sin(\frac{1}{2}\theta)},$$

where we have used that $\sin(\frac{1}{2}\theta) > 0$ for $0 < \theta < 2\pi$ (so no modulus is needed in the denominator) and that $|\sin x| \leq 1$. This is the required inequality, with equality exactly when $\sin((n+\frac{1}{2})\theta) = \pm 1$.

**Ask yourself** Why is the positivity of $\sin(\frac{1}{2}\theta)$ important?

If $\theta$ were allowed to range outside $(0, 2\pi)$, $\sin(\frac{1}{2}\theta)$ could be negative and the right-hand side of the inequality would be negative, which is impossible for a modulus. In the given range, $\frac{1}{2}\theta \in (0, \pi)$, so the bound is a genuine positive number.

## Checks

- $n = 1$: $\mathrm{C}_1 = 1 + \cos\theta = 2\cos^2(\frac{1}{2}\theta)$, and the formula gives $\cos(\frac{1}{2}\theta)\sin\theta / \sin(\frac{1}{2}\theta) = 2\cos^2(\frac{1}{2}\theta)$. Similarly $\mathrm{S}_1 = \sin\theta$ matches $\sin(\frac{1}{2}\theta)\sin\theta / \sin(\frac{1}{2}\theta)$.
- As $\theta \to 0$, the formula for $\mathrm{C}_n$ tends to $(n+1)$, which is the sum of $n+1$ ones; the bound $1/(2\sin\frac{1}{2}\theta)$ blows up, as it must.
- Numerically, with $n = 3$, $\theta = 1$: the direct sum is $\mathrm{C}_3(1) \approx 0.1342$, matching the formula, and $|\mathrm{C}_3 - \frac{1}{2}| \approx 0.366 \leq 1/(2\sin\frac{1}{2}) \approx 1.043$. Equality occurs, for example, at $n = 2$, $\theta = \pi/5$, where both sides equal $1.618\ldots$.

## Common slips

- Forgetting that the sums start at $k = 0$, so there are $n + 1$ terms: the geometric series has $e^{\mathrm{i}(n+1)\theta}$ on top, not $e^{\mathrm{i}n\theta}$.
- Taking real parts of $\left(e^{\mathrm{i}(n+1)\theta} - 1\right) / \left(e^{\mathrm{i}\theta} - 1\right)$ without first making the denominator real; the half-angle factorisation does this cleanly.
- Writing $|\sin(\frac{1}{2}\theta)|$ and not saying why it equals $\sin(\frac{1}{2}\theta)$.

## Beyond STEP

The function $\mathrm{C}_n(\theta) - \frac{1}{2} = \sin\left((n + \frac{1}{2})\theta\right) / \left(2 \sin \frac{1}{2}\theta\right)$ is, up to a factor, the *Dirichlet kernel*, the central object in the theory of Fourier series: the $n$th partial Fourier sum of a function is its average against this kernel. The bound proved here, together with the oscillation of the numerator, is the starting point for proving when Fourier series converge (and for Dirichlet's test for series such as $\sum \cos(k\theta)/k$). Abbott's *Understanding Analysis* has a readable chapter on Fourier series that uses exactly these sums.

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